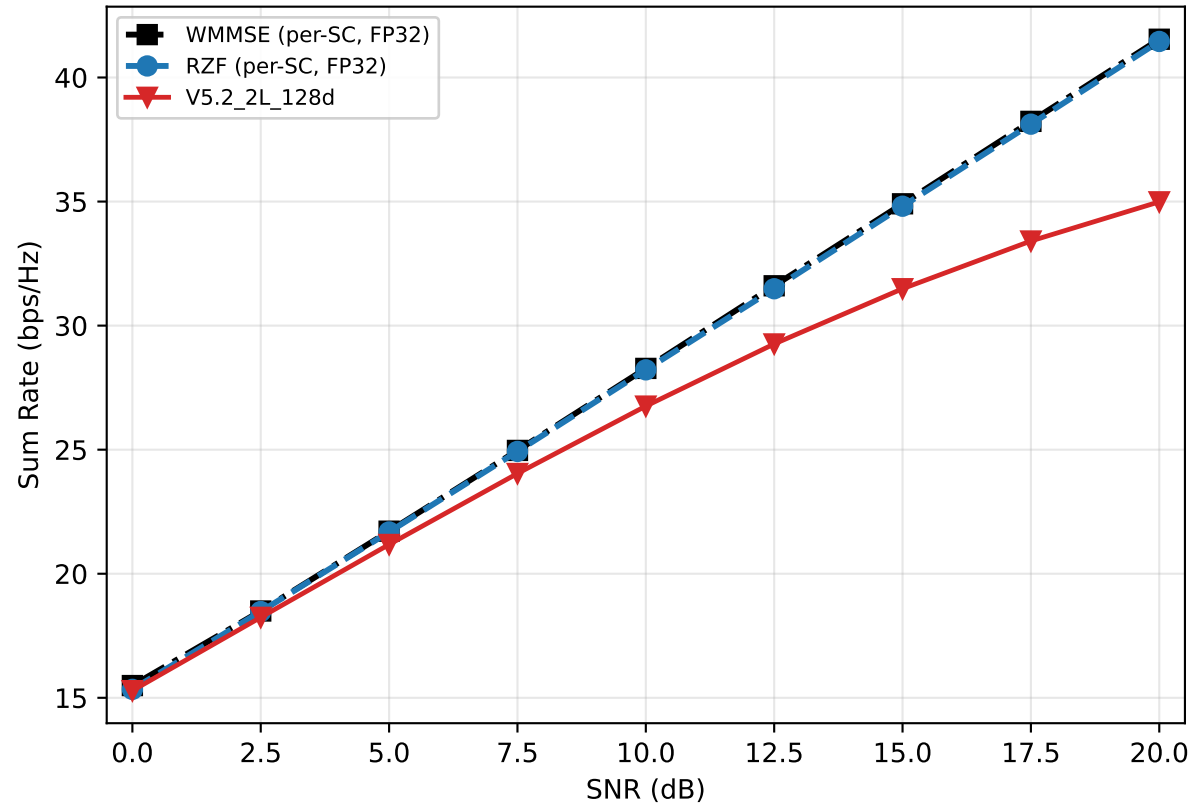
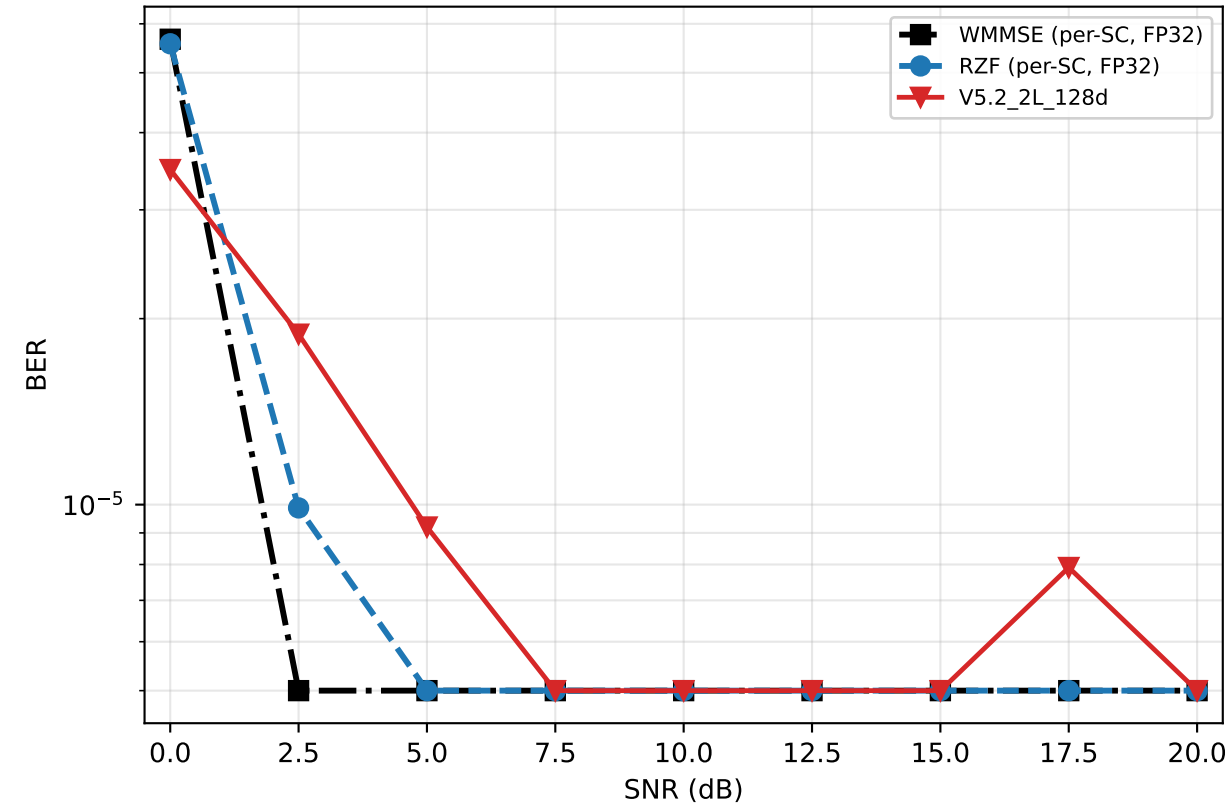


V4.2 Transformer Precoder — 8×4 MU-MIMO (8 TX, 4 UE, 72 SC, SNR train = 15 dB)
Ablation study: tokens per Resource Block (RB = 12 SC)

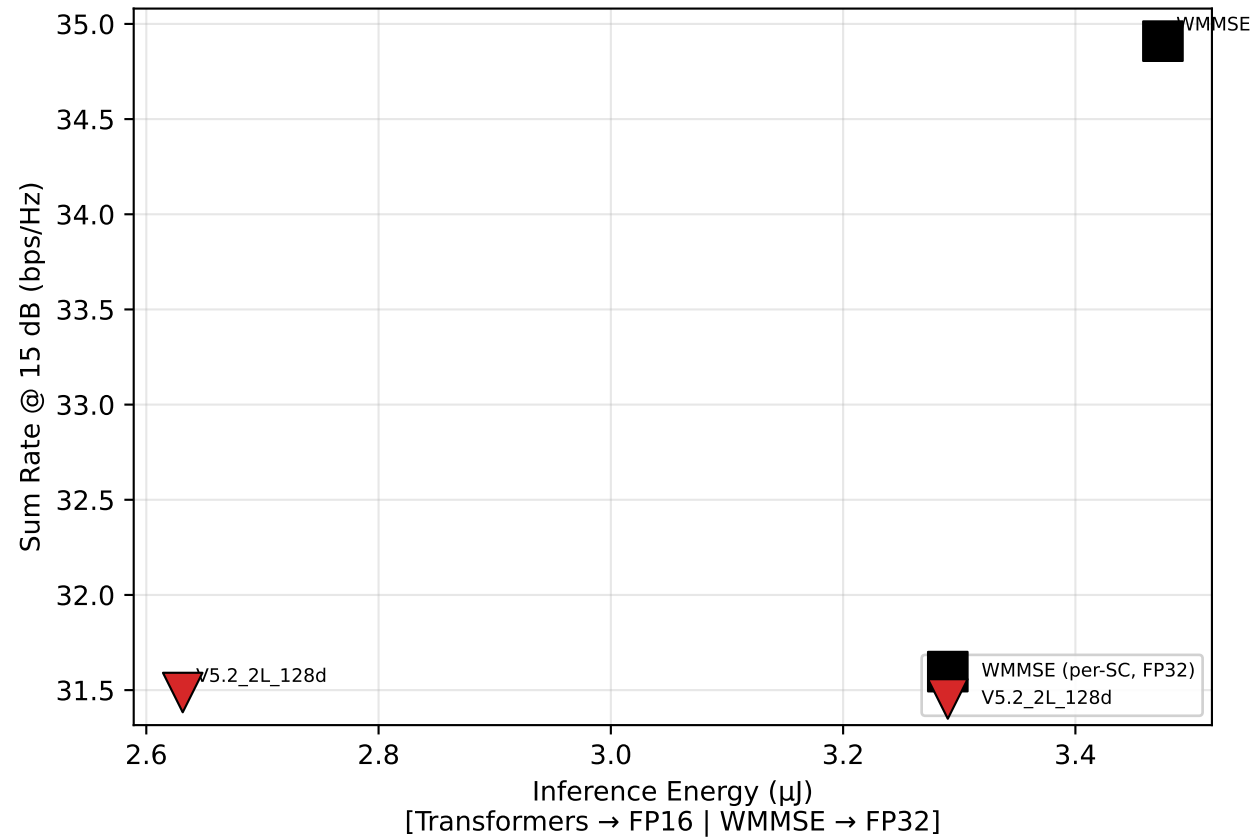
(a) Spectral Efficiency



(b) Bit Error Rate (Monte Carlo accumulation)



(c) Energy-Performance Pareto
(RZF excluded — see Fig. 2)



(d) Pareto Summary @ 15 dB
Energy gain = WMMSE_FP32 / model_FP16

Method	Rate (bps/Hz)	Gap vs WMMSE	FLOPs (M)	Energy FP16 (μ J)	Gain vs WMMSE (FP32)
WMMSE (per-SC, FP32)	34.91	+0.00	1842	N/A	ref
RZF (per-SC, FP32)	34.81	-0.10	2	N/A	ref
V5.2_2L_128d	31.49	-3.42	4853	2.6314	1.3×